

CSE 4621: Machine Learning Probability, Statistics and Datasets

Lecture 2

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Lecture Outline

- Dealing with Datasets
 - Types and formats
 - Total workflow
 - Dataset splits
- Probability Foundations
 - Probability distributions
 - Bayes' theorem
 - Maximum Likelihood Estimation



What is a Dataset?

A dataset is a structured collection of observations used to train, validate, test, or analyze models.

Hours Studied	Attendance	Current GPA	Final Grade
5	90%	3.4	A
2	60%	3.0	C

What is a Dataset?

- Sample → A single data point
- Feature → Input variables
- Label → Ground-truth output

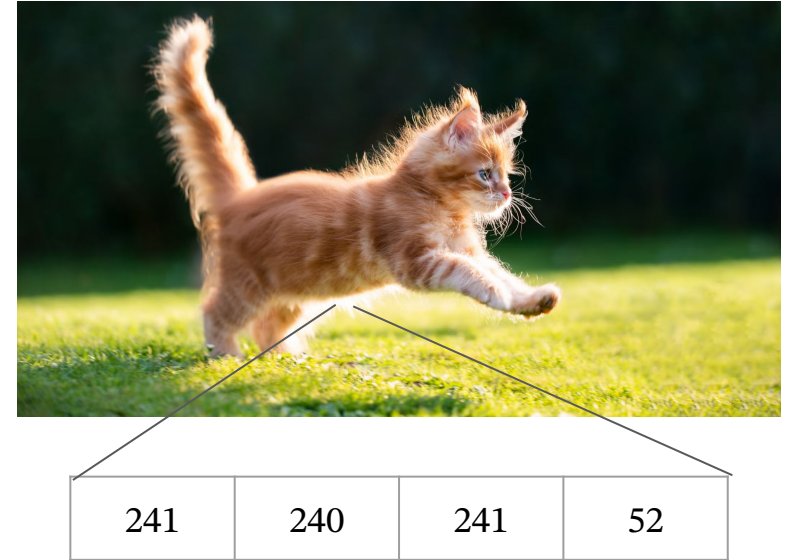
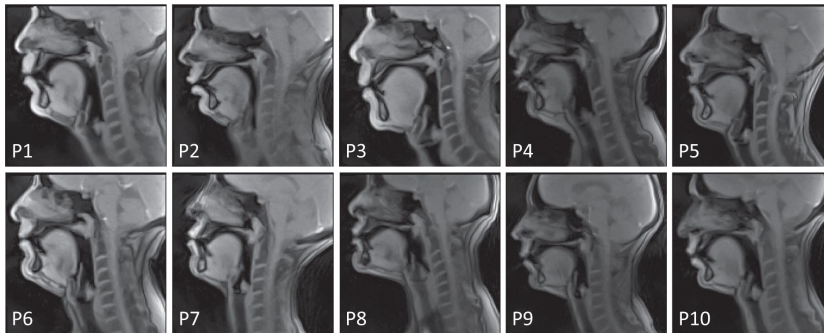
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What is a Dataset?

For images each image is a sample.

For features:

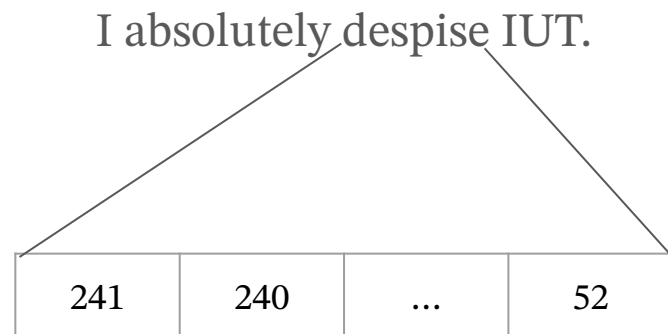
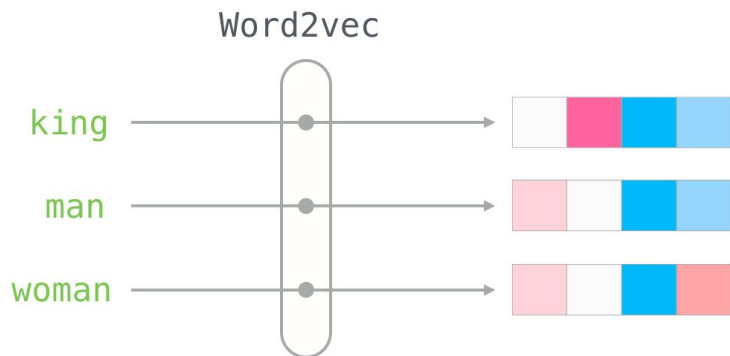
- All the pixels can be features
- Or some feature engineering/representation can be done



What is a Dataset?

For textual data:

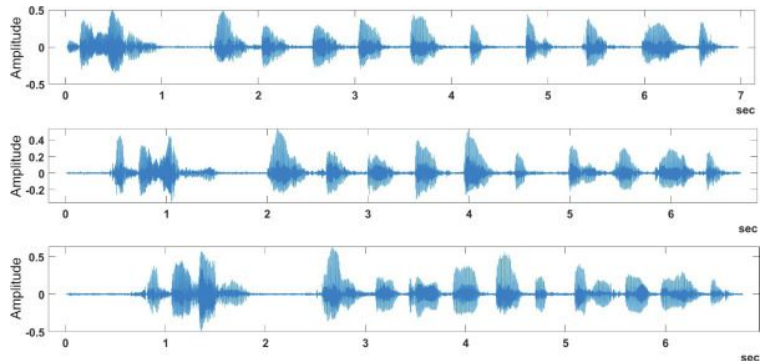
- Each word/sentence needs to be transformed into numbers
- Usually done while keeping semantic meaning



What is a Dataset?

For audio data:

- Usually represented as waveforms/spectrograms
- Frequency, amplitude, spectrogram patterns



Recommended Resources: [1]

Visual Stimuli



Our Reconstructed Video



SCREENSHOT FROM PAPER



What is a Dataset?

ML begins by converting messy reality into a representation a model can process

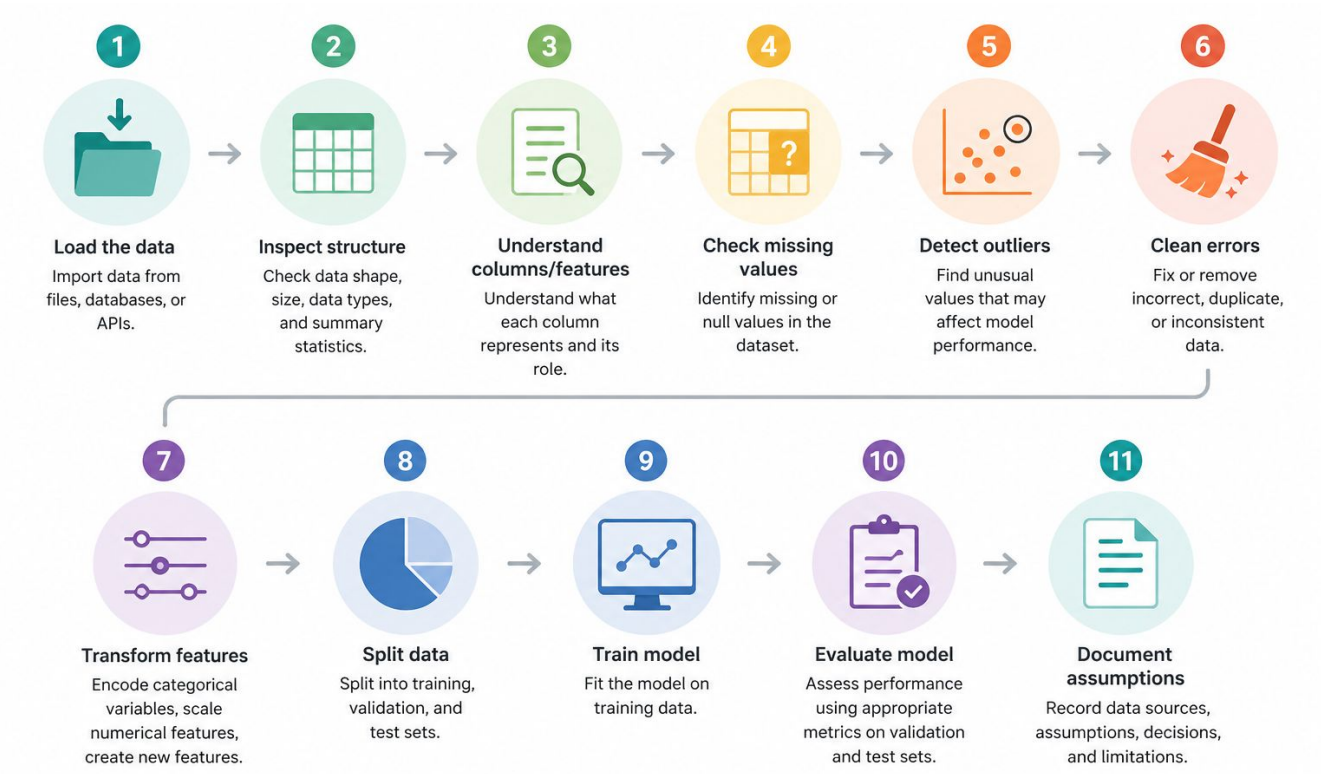
Supervised Datasets → Has Labels

Unsupervised Datasets → No Labels

Self-supervised Datasets → Labels are created from the data itself

Whenever accessing a new dataset, it is imperative to try some exploratory data analysis on it

How to Work with a Dataset?



Dataset Splitting

A model should be tested on data it has not seen. Thus, every dataset should be split:

- Training set → Used to learn parameters (70%)
- Validation set → Used to tune choices* (15%)
- Test set → Used for evaluation (15%)

If you keep checking the answer while preparing for a test, your performance is no longer reflecting your actual ability

- One train-validation-test split may get lucky or unlucky → Use multiple such splits (Cross Validation)

Data leakage

When information from the test set accidentally enters training:

- Scaling before splitting
- Duplicate samples across train and test
- Future information used to predict the past

Generalization

- A model's ability to perform well on new, unseen data.
- Bad model: Memorizes training examples.
- Good model: Learns reusable patterns.

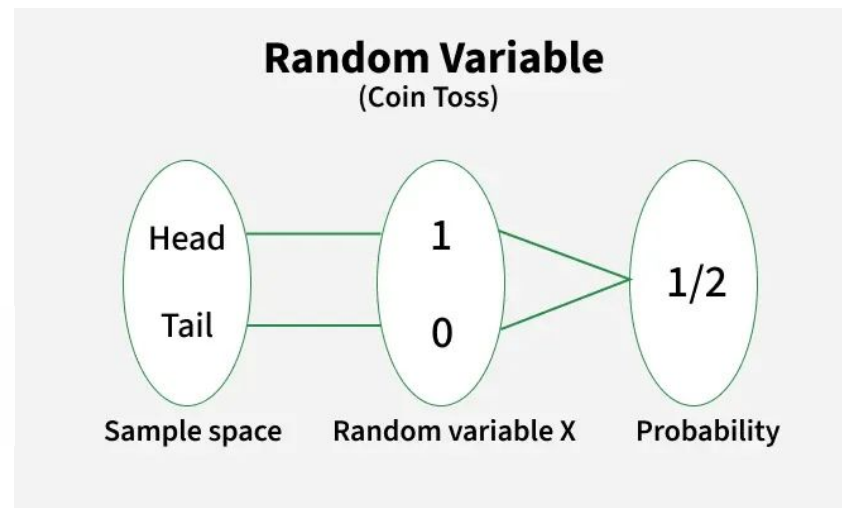
Probability Foundations

A **random variable** represents an uncertain quantity:

- X = height of a randomly selected student
- Y = whether an email is spam
- Z = # of customers arriving in one hour

Can either be *discrete* or *continuous*

$$\mathbb{E}[X] = \int x p(x) dx, \quad \text{Var}(X) = \mathbb{E}[(X - \mu)^2]$$



Bayes' Theorem

Probability of A given that B has happened:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

How to change assumptions based on new evidence?

$$P(H|D) = \frac{P(D|H)P(H)}{P(D)}$$

$H \rightarrow$ hypothesis , $D \rightarrow$ observed data , $P(H) \rightarrow$ prior belief

$P(D|H) \rightarrow$ likelihood, $P(H|D) \rightarrow$ posterior belief

Maximum Likelihood Estimation

You flip a coin 10 times and get 7 heads. What kind of coin would most likely produce this result?

- $p = 0.5$?
- $p = 0.7$?

Pick the explanation (probability) that makes your observed data most likely

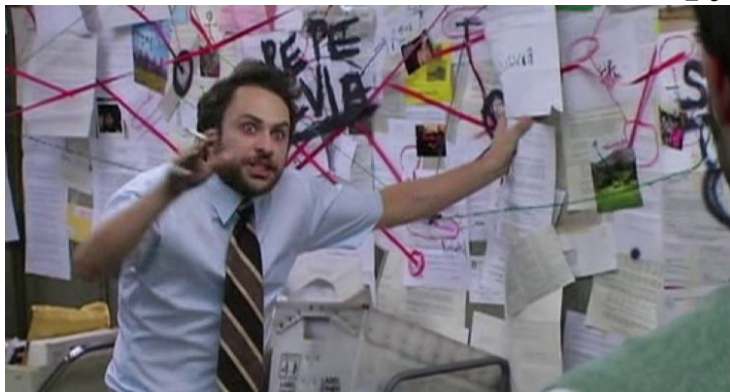
$$L(\theta) = P(data|\theta)$$

The goal is: $\theta_{MLE} = \operatorname{argmax} P(D|\theta)$

Maximum Likelihood Estimation

Many ML models are MLE in disguise:

- Linear regression with Gaussian noise $\rightarrow$ minimize Mean Squared Error
- Logistic regression $\rightarrow$ maximize Bernoulli likelihood
- Softmax classifier $\rightarrow$ maximize categorical likelihood
- Neural network classification $\rightarrow$ minimize cross-entropy



Textbook Chapters

- [Goodfellow] - Chapter 3
- [Bishop] - Chapter 2,
- [Alpaydin] - N/A

Recommended Resources

1. [Can AI read minds?](#)
2. [The Hitchhiker's Guide to Hyperparameter Tuning](#)

Thank you